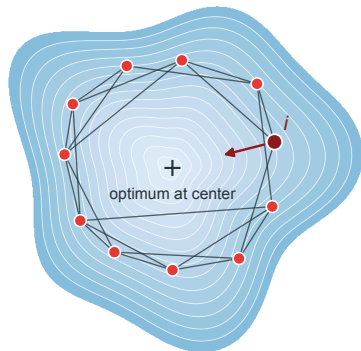
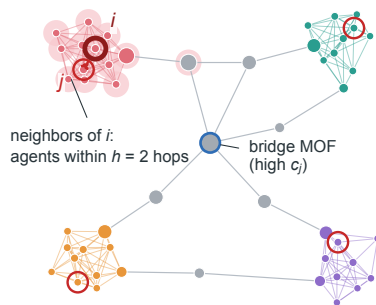


**a** SOCIAL (previous work)

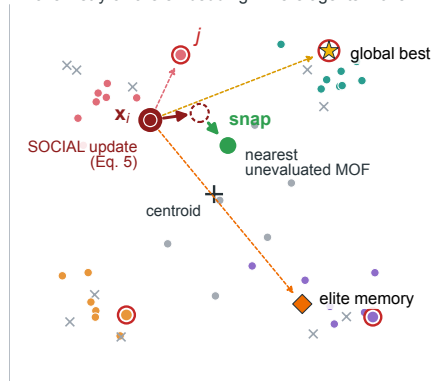
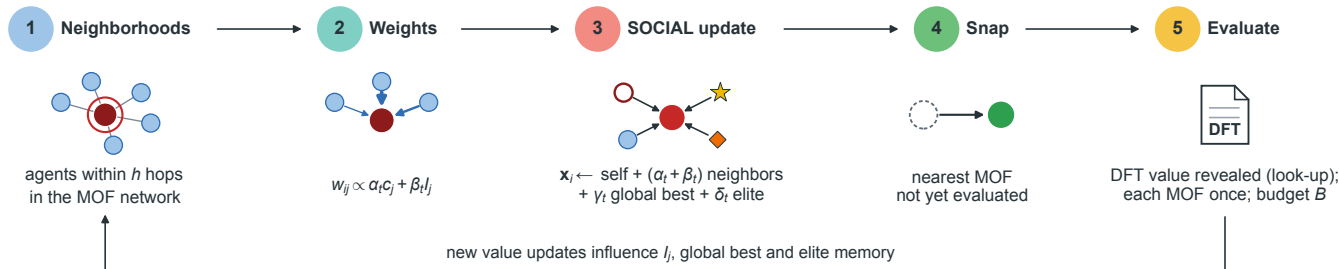
continuous search space,  
synthetic small-world agent graph

**b** SOCIAL-MGN (this study): the same agents in two views

MOF similarity network: who listens to whom



chemistry-aware embedding: where agents move

**c** One SOCIAL-MGN iteration (SOCIAL learning rules unchanged)

$\alpha_t, \beta_t$  decrease and  $\gamma_t, \delta_t$  increase over iterations; worse-than-median agents may jump to the least-explored chemical community